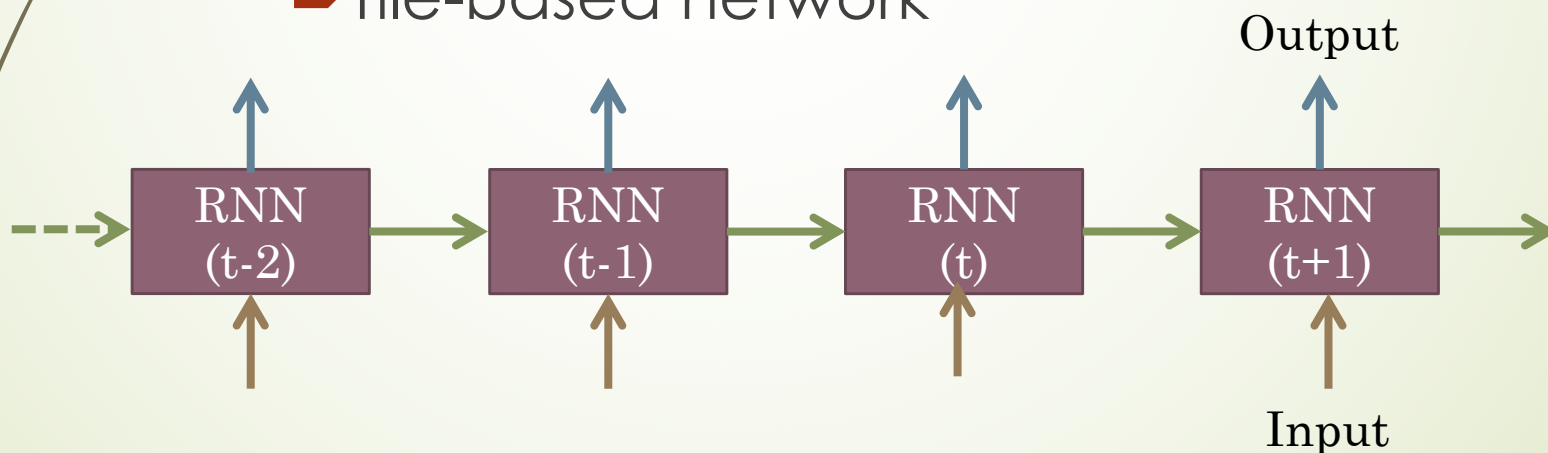
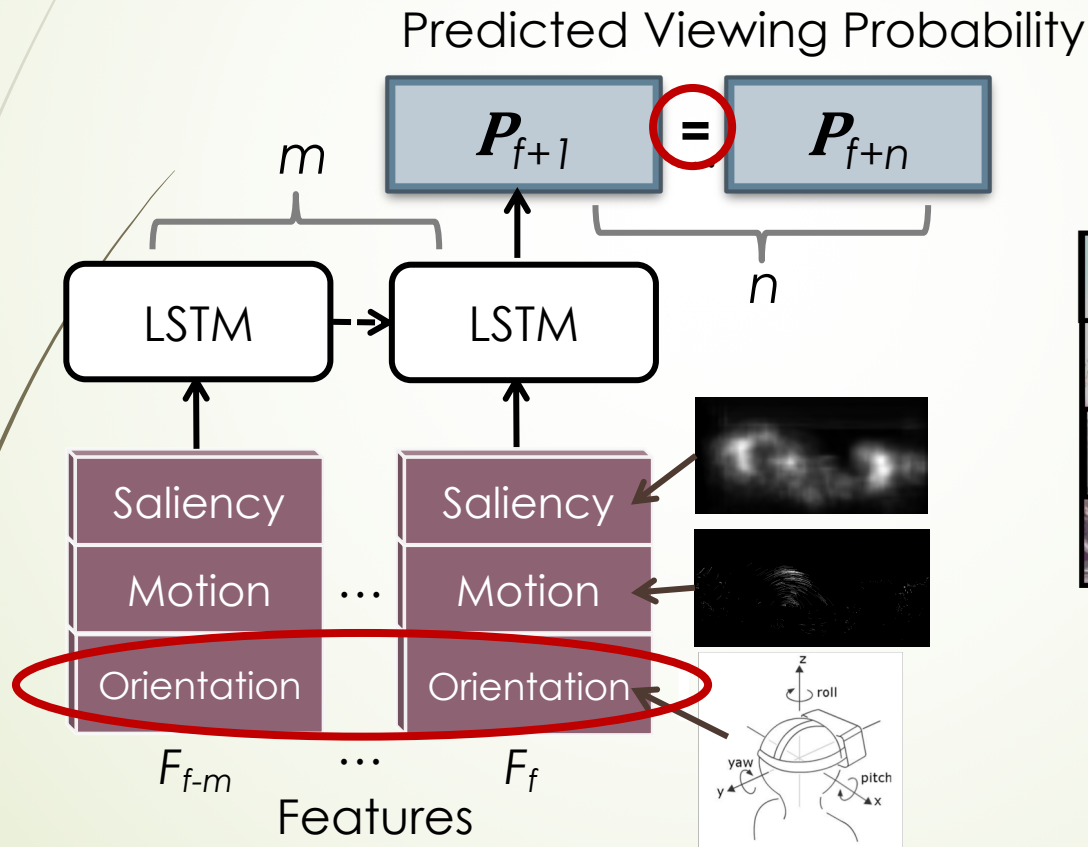


Recurrent Neural Networks

- Goal: predict the viewing probability of each tile in the next few seconds
 - Orientation-based network
 - Tile-based network



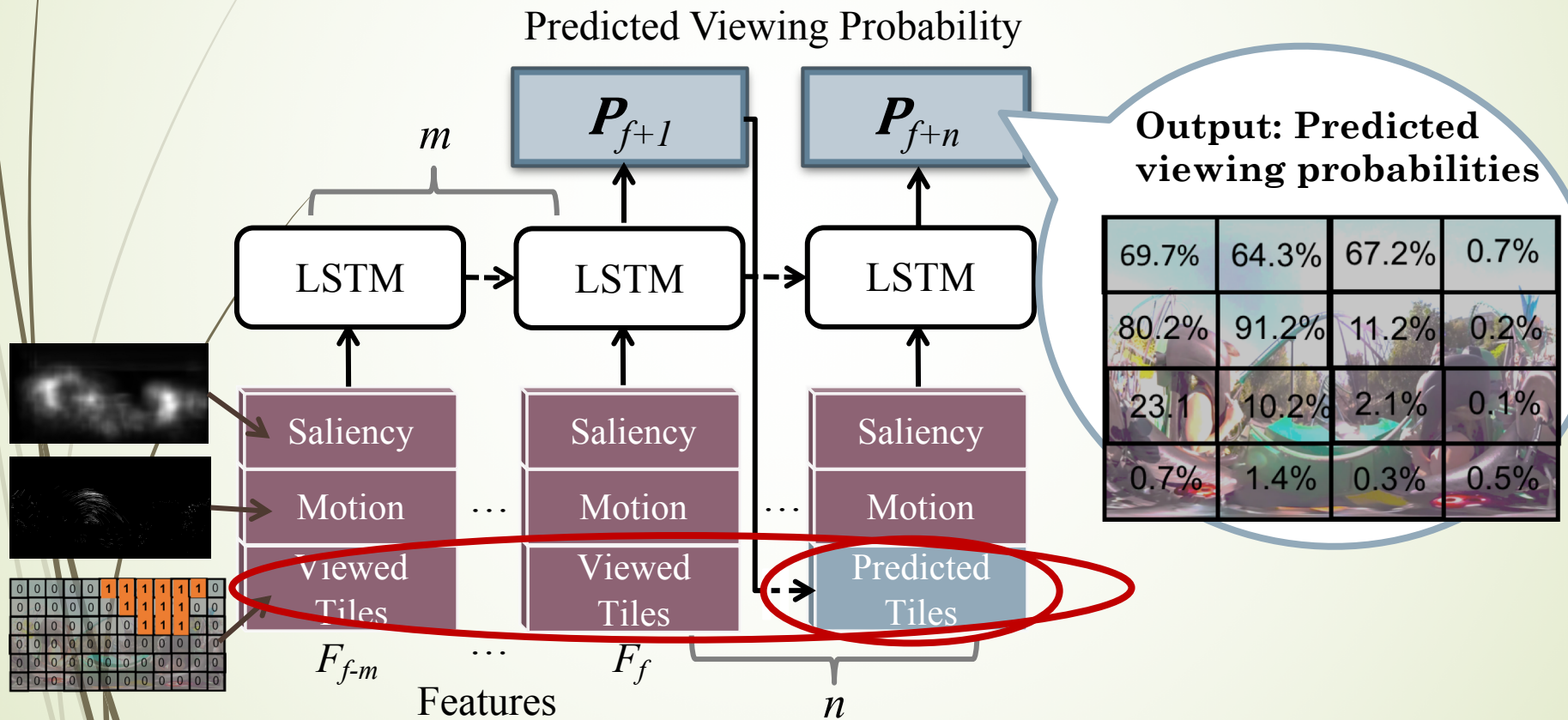
Orientation-Based Network



Output: Predicted viewing probabilities

69.7%	64.3%	67.2%	0.7%
80.2%	91.2%	11.2%	0.2%
23.1	10.2%	2.1%	0.1%
0.7%	1.4%	0.3%	0.5%

Tile-Based Network



Training Results

► Orientation-based network

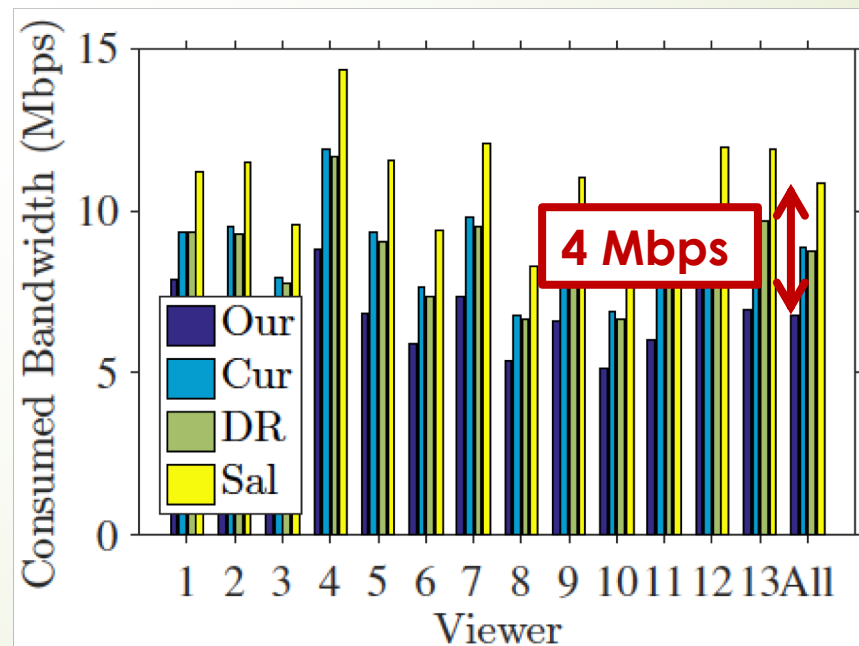
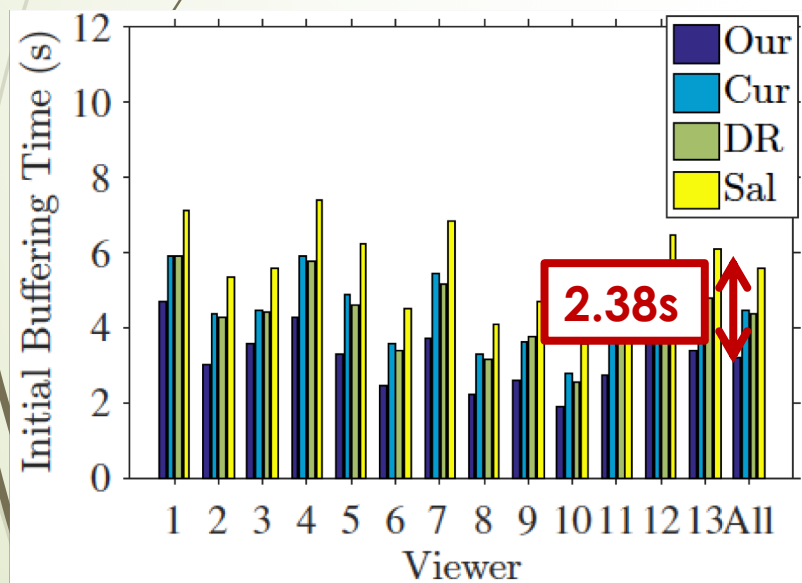
Parameters			Training Set			Testing Set		
No. Neu.	LSTM Layers	Drop.	Rank. Loss	Accuracy	F-score	Rank. Loss	Accuracy	F-score
256	1	T	0.1	88.20%	0.67	0.15	85.72%	0.60
512	1	T	0.09	89.25%	0.70	0.14	86.35%	0.62
1024	1	T	0.09	89.28%	0.71	0.14	86.06%	0.62

► Tile-based network

Parameters			Training Set			Testing Set		
No. Neu.	LSTM Layers	Drop.	Rank. Loss	Accuracy	F-score	Rank. Loss	Accuracy	F-score
256	2	F	0.14	86.58%	0.57	0.20	83.94%	0.52
512	2	F	0.13	86.91%	0.58	0.19	84.11%	0.52
1024	2	F	0.12	87.29%	0.60	0.19	84.22%	0.53

Sample Results

- Configure the streaming server to ensure $< 10\%$ missing tile ratio
- Shorter initial buffering time
- Smaller bandwidth consumption



Agenda

- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges